

SANTIAGO QUINTA NORMAL, Chile

WMO: 855770

Lat: 33.4450S

Lon: 70.6828W

Elev: 1706

StdP: 13.81

Time Zone: -4.00 (CHL)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	34.8	26.3	21.5	58.8	28.8	24.3	52.1	7.1	52.3	5.6	52.8	0.0	N/A	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.6	90.0	62.7	87.8	62.4	85.8	62.1	65.9	83.8	64.8	82.9	63.7	81.8	6.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.6	78.3	72.1	57.6	75.5	70.9	56.5	72.5	69.8	31.5	83.9	30.6	83.0	29.8	82.0	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	71.9	71.3	67.8	60.0	54.8	49.4	48.7	51.3	56.0	60.2	65.3	69.0
	DBStd	9.11	2.81	3.38	3.44	4.35	4.17	4.07	4.42	4.74	4.76	4.78	4.17	3.63
	HDD50	188	0	0	0	0	9	59	78	38	4	0	0	0
	HDD65	2375	0	3	13	157	316	469	505	425	271	161	46	9
	CDD50	3994	679	595	552	300	158	41	38	79	185	317	460	590
	CDD65	703	214	179	100	7	0	0	0	0	1	12	56	134
	CDH74	8398	2213	1748	1215	233	11	0	2	26	83	302	875	1690
	CDH80	2652	818	594	340	36	0	0	0	4	8	57	246	549
(14) Wind	WSAvg	2.1	3.4	3.0	2.4	1.7	1.1	0.8	1.0	1.3	2.0	2.5	3.0	3.3
(15) Precipitation	PrecAvg	10.2	0.0	0.0	0.1	0.4	1.5	2.8	2.2	1.5	0.7	0.4	0.2	0.1
	PrecMax	23.1	0.3	1.0	1.0	3.9	6.7	9.0	9.7	6.3	4.3	2.1	2.6	1.7
	PrecMin	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.2	0.0	0.2	0.2	0.7	1.5	2.5	2.1	1.4	0.9	0.5	0.5	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.0	91.5	90.1	84.0	75.9	70.4	72.3	78.1	80.8	86.0	90.3	91.6
		MCWB	63.9	62.7	61.9	59.0	56.5	53.1	54.8	56.7	57.3	61.3	60.9	63.2
	2%	DB	89.9	88.7	86.7	80.2	71.4	66.6	66.8	71.8	76.2	81.0	85.9	88.5
		MCWB	63.4	63.0	61.4	58.3	55.1	52.3	52.4	54.0	56.9	58.1	60.2	62.0
	5%	DB	87.4	86.4	83.8	76.5	67.8	62.8	62.2	66.4	72.5	77.2	82.9	85.5
		MCWB	63.3	62.7	61.1	57.5	54.2	50.9	50.4	52.7	56.0	57.4	59.7	61.7
	10%	DB	84.8	83.9	81.1	72.9	64.5	58.9	58.6	61.7	67.5	73.4	79.5	83.1
		MCWB	63.1	62.7	60.6	56.9	53.7	49.8	49.1	50.9	53.9	56.3	59.1	61.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	67.4	65.4	61.7	59.7	56.2	56.1	57.8	60.3	63.1	63.8	65.7
		MCDB	86.3	85.0	82.8	77.3	67.2	65.8	68.4	74.6	76.7	80.7	82.6	84.4
	2%	WB	66.1	65.9	63.7	60.1	57.4	54.0	53.6	55.2	58.2	60.1	62.1	64.2
		MCDB	84.5	83.1	81.4	75.2	65.7	62.2	62.7	68.3	72.9	76.4	81.1	83.5
	5%	WB	64.8	64.7	62.5	58.8	56.0	52.2	51.9	53.3	56.5	58.5	60.9	63.0
		MCDB	83.5	81.7	79.6	72.7	64.4	59.2	59.0	64.2	69.8	74.1	79.6	82.4
	10%	WB	63.8	63.4	61.4	57.4	54.7	50.8	50.3	51.6	54.8	56.9	59.6	61.9
		MCDB	82.1	80.6	78.2	70.7	62.4	57.3	56.8	60.2	65.6	71.4	77.3	80.8
(35) Mean Daily Temperature Range	5% DB	MDBR	26.6	26.4	27.2	23.9	19.6	19.1	19.4	20.2	21.1	23.6	26.1	27.9
		MCDBR	30.7	29.6	31.3	31.1	26.5	27.0	27.0	29.9	30.8	31.3	32.6	31.4
		MCWBR	10.5	10.1	10.7	12.0	11.6	13.8	13.5	13.9	12.7	11.8	11.7	11.5
	5% WB	MCDBR	26.3	25.5	27.9	26.4	20.6	21.4	22.6	25.2	26.9	27.3	28.7	28.7
		MCWBR	10.1	9.7	10.5	11.1	9.7	11.8	11.8	12.2	11.7	11.2	11.4	11.4
(40) Clear-Sky Solar Irradiance	taub		0.345	0.344	0.328	0.324	0.315	0.302	0.304	0.318	0.338	0.329	0.320	0.332
	taud		2.407	2.423	2.458	2.461	2.484	2.546	2.502	2.451	2.373	2.409	2.447	2.429
	Ebn at Noon		315	309	302	285	271	267	272	283	293	310	321	320
	Edh at Noon		40	38	34	30	26	23	25	30	36	38	38	39
(44) All-Sky Solar Radiation	RadAvg		2737	2466	2018	1400	891	674	740	963	1397	1886	2417	2738
	RadStd		128	88	107	141	105	97	80	121	148	146	148	99

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.68	N/A	-2.11	+1.04	N/A	N/A	N/A	N/A	N/A	+247	+167

Nomenclature: See separate page